Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree Examination June 2022 (2019 scheme)

Course Code: EET204

Course Name: ELECTROMAGNETIC THEORY

Max. Marks: 100 Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

			Marks
1		Two vectors $\boldsymbol{A} = 4\boldsymbol{ax} + 8\boldsymbol{ay}$ and $\boldsymbol{B} = 12\boldsymbol{ay} - 8\boldsymbol{az}$ are on the same plane. Find the vector normal to the plane.	3
2		Develop the equation for differential volume dv in Spherical coordinate system. Draw necessary figure.	3
3		State Coulomb's law and write the equation in vector form.	3
4		What are equipotential surfaces. Mention two characteristics of equipotential surfaces.	3
5		Derive the expression for $\boldsymbol{H}$ at a point due an infinitely long conductor carrying a current I.Use Biot Savart's law.	3
6		State Ampere's circuital law for steady magnetic fields. Write the point form of Ampere's circuital law.	3
7		Describe the characteristics of uniform plane waves.	3
8		What is meant by a Poynting vector. Mention its significance.	3
9		What are the primary constants of Transmission line. Draw a neat figure to represent it.	3
10		Define characteristic impedance of a lossless line.	3

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11	a)	Find the Vector directed from (2, -5, -2) to (14, -5, 4) in Spherical coordinate system.	7
	b)	Define Gradient of a Vector and Mention three characteristics of it.	7
12	a)	State Divergence of a vector field and Derive the expression for Divergence of a vector field in integral and point forms.	7
	b)	Derive Coordinate transformation between Cartesian to Cylindrical Coordinates with necessary figure.	7

Module -2

13	a)	State Gauss's law and obtain Electric field intensity E at any point due to an infinite sheet of charge with the help Gauss's law. Draw necessary figures.	7
	b)	Obtain the Potential at a point (1,2,5) due to a dipole with centre at origin and of dipole moment **P** = 20Cm acting along the Z axis. Also find the Potential at point (1,2,0).	7
14	a)	Find the Voltage between two infinitely long conductors parallel to z axis with charges +4nC and -4nC are passing through points (0,-4,0) and (0,4,0) respectively. Assume free space.	7
	b)	Derive the expression for Capacitance per unit length between two infinitely long conductors.	7

Module -3

15	a)	List all the Maxwell's equation for static Electric and Magnetic fields in Integral and Point forms. Write the significance of each in one sentence.	7
	b)	An uniform electric field $\boldsymbol{E_1} = 5ax - 2ay + 4az$ incidents at the boundary between two dielectrics of permittivity $\epsilon_1 = 2$ and $\epsilon_2 = 4$ respectively. Find the Vector $\boldsymbol{E_2}$. Assume the boundary is at Z=0 plane.	7
16	a)	Derive the expression for modified form of Ampere's circuital law.	7
	b)	The magentic flux density at a point which is at a perpendicular distance of 2m from an infinitely long current carrying conductor along Z axis is 0.05 mWb/sq.m. Find the value of dc current passing through the conductor. Also find the value of B at point (1, 2, 5) during the same condition as above. Assume free space.	7

Module -4

17		Derive Poynting Theorem. Describe each term in the equation with one sentence.	14
18		Calculate the intrinsic impedance η, the propagation constant γ, and the wave velocity u at a frequency f = 50 Hz for a good conducting medium in which σ = 58 S/m and $\mu r = 1$.	14

Module -5

19		Describe the terms; a) SWR. b) Impedance matching c) Propagation constants.	14
20		Derive the wave equations for Voltage and Current in a transmission line with neat figure.	14

*****